

Public Bus Live Tracking & Crowding Estimator

Complete Requirements Table | DBMS Experiential Learning – Level 3

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	System shall ingest GPS coordinates from buses and calculate ETA for upcoming stops.	High	ETA updates using current bus location, speed, and distance.	Provides accurate arrival information to commuters.
FR-002	Functional	System shall allow commuters to view the live location of buses on a selected route.	High	Current bus position is displayed and refreshed using the latest GPS data.	Helps commuters track approaching buses.
FR-003	Functional	System shall display estimated arrival time of selected buses at stops.	High	ETA is displayed for the selected stop and updates when new data arrives.	Helps commuters plan their journey.
FR-004	Functional	System shall estimate bus crowding using passenger/ticketing sensor data.	High	System categorizes crowding as low, medium, or high based on passenger data.	Helps commuters avoid overcrowded buses.
FR-005	Functional	System shall notify users about significant delays or overcrowding.	Medium	Alert is generated when configured delay or crowding thresholds are exceeded.	Keeps commuters informed about disruptions.
NFR-001	Performance & Security	System shall handle telemetry from up to 1,000 active transit vehicles simultaneously.	High	Benchmarking confirms target latency and security standards under peak load.	Ensures the system works at city scale securely.
NFR-002	Availability & Reliability	System shall maintain tracking availability and provide the latest valid information during temporary data/network interruptions.	High	System recovers from interruptions and displays the latest valid data.	Prevents loss of useful tracking information.

Requirement count: Exactly 5 Functional Requirements (FR-001 to FR-005) and 2 Non-Functional Requirements (NFR-001 to NFR-002).